ASSIGNMENT 9: Apply Object Detection from CCTV Footage

1. Aim

To implement a basic object detection system using Python and OpenCV to identify people in CCTV-style video footage and display bounding boxes around the detected individuals.

2. Objectives

1. To understand the concept of computer vision.
2. To learn how to read video frames using OpenCV.
3. To implement a pre-trained people detector.
4. To identify people in video frames.
5. To draw bounding boxes around detected people.
6. To display the processed video in real time or frame by frame.

3. Theory

Computer vision is a field of artificial intelligence that enables computers to analyse images and videos and extract useful information from them.

Object detection is a computer vision task that identifies objects and estimates their locations within an image. A detection system commonly provides a class label and a bounding box for each detected object.

A video consists of a sequence of frames. An object detection program reads the frames one by one and processes them to locate objects.

CCTV object detection can be used in occupancy monitoring, pedestrian analysis, traffic monitoring, and safety-related research.

This assignment uses OpenCV's built-in Histogram of Oriented Gradients (HOG) people detector. HOG represents image patterns using gradient information, and a pre-trained detector uses those patterns to recognise shapes resembling people.

The detector used here focuses on people rather than all possible object categories. More advanced detectors can recognise vehicles, bags, animals, and other categories.

The detection results depend on video quality, lighting, distance, camera angle, and whether people are partially hidden. Therefore, this program is a demonstration rather than a complete security system.

4. Algorithm

1. Start the program.
2. Import the OpenCV library.
3. Open the CCTV video file.
4. Check whether the video is accessible.
5. Initialise the HOG people detector.
6. Read a video frame.
7. Resize the frame for processing.
8. Detect people in the frame.
9. Draw bounding boxes around detected people.
10. Add a label to each bounding box.
11. Display the processed frame.
12. Continue until the video ends or the user presses `q`.
13. Release the video and close the display window.

5. Explanation of the code

1. `cv2.VideoCapture()` opens the video file.
2. `cap.isOpened()` checks whether the video was opened successfully.
3. `cv2.HOGDescriptor()` creates the HOG descriptor.
4. `getDefaultPeopleDetector()` loads OpenCV's built-in people detector.
5. `cap.read()` reads the next video frame.
6. `cv2.resize()` resizes the frame to 640 × 360 pixels.
7. `detectMultiScale()` searches for people in the frame and returns bounding boxes.
8. `cv2.rectangle()` draws a rectangle around each detected person.
9. `cv2.putText()` displays the label `Person`.
10. `cv2.imshow()` displays the processed video frame.

11. `cap.release()` releases the video resource.
12. `cv2.destroyAllWindows()` closes the OpenCV windows.

6. Procedure to Run the Program in VS Code

Step 1: Install OpenCV in your Python environment.

Step 2: Create a folder for Assignment 9.

Step 3: Save the Python program as `assignment9.py`.

Step 4: Place an authorised video named `cctv.mp4` in the same folder.

Step 5: Open the folder in VS Code and open the integrated terminal.

Step 6: Run the program.

Step 7: The video window will display the frames with bounding boxes around detected people.

Step 8: Press `q` to exit.

7. Applications

1. Pedestrian detection.
2. Occupancy monitoring demonstrations.
3. Traffic analysis.
4. Video analytics research.
5. People-counting prototypes.
6. Computer vision education and experimentation.

8. Advantages

1. Demonstrates a practical computer vision application.
2. Uses an existing detector provided by OpenCV.
3. Processes video frame by frame
4. Shows detected people visually using bounding boxes.
5. Requires no separate model download.

9. Limitations

1. HOG detection is not always accurate.
2. It may fail to detect partially hidden people.
3. Processing speed depends on video resolution and computer hardware.

10. Conclusion

A basic CCTV person-detection system was implemented using Python and OpenCV. The program reads video frames, detects people using the HOG detector, and draws bounding boxes around recognised individuals. This experiment demonstrates the basic working of object detection in computer vision.